# Olist Brazilian E-Commerce
# Data Analysis Project Report


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| **Tools** | PostgreSQL, Power BI Desktop |
| **Dataset** | Olist Brazilian E-Commerce (Kaggle, 2016–2018) |
| **Date** | May 2026 |


**Abstract.** This report documents the findings of a comprehensive data analysis performed on the Olist Brazilian E-Commerce public dataset. The analysis covers 99,440 orders placed between 2016 and 2018 across Brazil. Using PostgreSQL for data querying (21 business questions across 11 analytical sections) and Power BI for visual reporting (4 dashboard pages), this project addresses key business areas: order fulfilment, customer behaviour, revenue performance, payment preferences, logistics efficiency, and seller analysis.


### Key Performance Indicators

| KPI | Value | KPI | Value |
|---|---|---|---|
| Total Orders | 99,440 | Total Revenue | R$ 13.59M |
| Delivered Orders | 96,480 | Total Customers | 96,100 |
| Undelivered Orders | 2,963 | Customer Loyalty Rate | 3.12% |
| Fulfilment Rate | 97.02% | Total Sellers | 3,095 |

## Contents

## 1. Dataset & Schema Overview

The dataset contains approximately 100,000 orders placed on the Olist marketplace between 2016 and 2018. It spans 10 relational tables covering customers, orders, products, sellers, payments, and reviews.

| Table | Rows | Description |
|---|---:|---|
| `customers` | 99,441 | Customer IDs, city, state, zip code |
| `orders` | 99,441 | Order lifecycle — status, timestamps |
| `order_items` | 112,650 | Product-level details: price, freight |
| `products` | 32,951 | Attributes, weight, dimensions, category |
| `sellers` | 3,095 | Seller city, state, zip code |
| `payments` | 103,886 | Payment type, installments, value |
| `reviews` | 100,000 | Review score (1–5), comments |
| `geolocation` | 1,000,163 | Lat/Lng coordinates by zip prefix |
| `product_category_translation` | 71 | Portuguese to English category mapping |
| `brazil_states` | 27 | State code to full state name (custom) |

**Note:** `customer_unique_id` was used throughout (not `customer_id`) to avoid double-counting customers with multiple orders. The `brazil_states` table was custom-created to improve readability across queries.

## 2. Order Analysis

### 2.1 Order Status Distribution

97.02% of all orders were successfully delivered, indicating a healthy fulfilment rate. The remaining 2,963 orders are spread across statuses: Shipped, Canceled, Unavailable, Invoiced, Processing, Created, and Approved. This undelivered segment represents an operational risk and should be monitored in real time.

### 2.2 Monthly Order Volume Trend

Order volume grew significantly from 2016 through mid-2018. Peak activity falls in Q1 and Q2, suggesting seasonal demand spikes possibly tied to Brazilian holidays and events. The sharp drop in late 2018 reflects the dataset cutoff, not an actual business decline.

### 2.3 Customer Ordering Pattern by Weekday

| Day | Orders |
|---|---:|
| Monday | 16,200 |
| Tuesday | 15,960 |
| Wednesday | 15,550 |
| Thursday | 14,760 |
| Friday | 14,120 |
| Saturday | 10,890 |
| Sunday | 11,960 |

**Recommendation:** Schedule marketing campaigns and flash sales on Monday and Tuesday to capture peak customer engagement.

## 3. Customer Analysis

### 3.1 Geographic Distribution (Top 5 States)

| State | Customers | Share |
|---|---|---|
| São Paulo | 40,300 | 42.0% |
| Rio de Janeiro | 12,380 | 13.0% |
| Minas Gerais | 11,260 | 11.7% |
| Rio Grande do Sul | 5,280 | 5.5% |
| Paraná | 4,880 | 5.1% |

**Finding:** São Paulo alone accounts for $\sim$42% of the entire customer base, making it the most critical market. This also represents a geographic concentration risk.

### 3.2 Customer Loyalty Rate

Only 3.12% of customers placed more than one order. This low repeat-purchase rate indicates that the vast majority of users are single-purchase customers.

**Recommendation:** Invest in loyalty programs, personalised post-purchase email campaigns, and re-engagement flows targeting high-monetary single-order customers identified via RFM segmentation (see Section 10).

## 4. Revenue Analysis

### 4.1 Revenue by State (Top 5)

| State | Revenue | Share |
|---|---|---|
| São Paulo | R$ 5.20M | 38.3% |
| Rio de Janeiro | R$ 1.82M | 13.4% |
| Minas Gerais | R$ 1.59M | 11.7% |
| Rio Grande do Sul | R$ 0.75M | 5.5% |
| Paraná | R$ 0.68M | 5.0% |
| **Total** | **R$ 13.59M** | **100%** |

### 4.2 Monthly Revenue Performance

| Month | Revenue | Note |
|---|---|---|
| January | R$ 1.04M | |
| February | R$ 1.06M | |
| March | R$ 1.31M | MoM +43.45% |
| April | R$ 1.31M | |
| May | R$ 1.47M | Peak month |
| June | R$ 1.28M | |
| July | R$ 1.35M | |
| August | R$ 1.39M | |
| September | R$ 0.61M | Seasonal trough; MoM −56.40% |
| October | R$ 0.69M | |
| November | R$ 0.99M | MoM +43.45% recovery |
| December | R$ 0.73M | |

**Finding:** Revenue peaked in May (R$ 1.47M) and crashed in September (−56.40% MoM), suggesting a pronounced seasonal trough mid-year.

## 5. Payment Analysis

| Payment Type | Orders | Share | Revenue |
|---|---|---|---|
| Credit Card | 77,000 | 75.24% | R$ 10.22M |
| Boleto | 20,000 | 19.46% | R$ 2.66M |
| Voucher | 4,000 | 3.80% | R$ 0.52M |
| Debit Card | 2,000 | 1.50% | R$ 0.19M |

**Finding:** Credit card dominates at 75.24%. Boleto (Brazilian bank slip) is a strong secondary channel at 19.46%, reflecting local payment behaviour and the unbanked population.

**Recommendation:** Optimise the credit card checkout UX and introduce instalment incentives to increase average order value.

## 6. Product & Category Analysis

### 6.1 Top-Selling Categories by Delivered Orders

| Category | Delivered Orders | Note |
|---|---|---|
| Bed, Bath & Table | 9,270 | Highest volume |
| Health & Beauty | 8,650 | Repeat-purchase potential |
| Sports & Leisure | 7,530 | Seasonal demand |
| Computers & Accessories | 6,530 | High-value items |
| Watches & Gifts | 5,500 | Gift season peaks |

### 6.2 Categories with Most Late Deliveries

| Category | Late Deliveries | Note |
|---|---|---|
| Bed, Bath & Table | 920 | |
| Health & Beauty | 857 | |
| Furniture & Decor | 688 | |
| Sports & Leisure | 625 | |
| Computers & Accessories | 594 | |

**Finding:** Top-selling categories also record the most late deliveries due to volume alone. Monitor late delivery *rate* (%) rather than absolute count to isolate genuine operational failures from volume effects.

## 7. Delivery & Logistics Analysis

### 7.1 States with Highest Average Delivery Time

| State | Avg Delivery Days |
|---|---|
| Roraima | 29.34 |
| Amapá | 27.18 |
| Amazonas | 26.36 |
| Alagoas | 24.50 |
| Pará | 23.73 |
| Maranhão | 21.51 |
| Sergipe | 21.46 |
| Ceará | 21.20 |
| Acre | 21.00 |
| Paraíba | 20.39 |

**Finding:** Northern and Northeastern states face 20–30 day delivery times due to geographic distance, poor road infrastructure, and limited logistics network coverage.

### 7.2 Customer Satisfaction vs. Delivery Time

| Review Score | Avg Delivery Days |
|---|---|
| 1 (Worst) | 21.25 |
| 2 | 16.60 |
| 3 | 14.21 |
| 4 | 12.25 |
| 5 (Best) | 10.62 |

**Finding:** There is a clear negative correlation between delivery time and review score. 5-star orders average 10.62 days vs. 21.25 days for 1-star orders. **Delivery speed is the single biggest driver of customer satisfaction.**

**Recommendation:** Prioritise logistics investment in Northern states and negotiate SLA improvements with regional carriers.

## 8. Seller Analysis

### 8.1 Top Revenue-Generating Sellers

| Seller ID | Revenue |
|---|---|
| 4869f7a5dfa277a7d... | R$ 0.23M |
| 53243585a1d6dc26... | R$ 0.22M |
| 4a3ca9315b744ce9... | R$ 0.20M |
| fa1c13f2614d7b5c... | R$ 0.19M |
| 7c67e1448b00f6e9... | R$ 0.19M |
| 7e93a43ef30c4f03... | R$ 0.18M |
| da8622b14eb17ae2... | R$ 0.16M |
| 7a67c85e85bb2ce8... | R$ 0.14M |
| 1025f0e2d44d7041... | R$ 0.14M |
| 955fee9216a65b61... | R$ 0.14M |

**Finding:** The top 10 sellers collectively generate ∼R$1.89M, roughly 14% of total revenue. This seller concentration is a supply-chain risk.

### 8.2 Seller Churn (2017 to 2018)

Several sellers who were active in 2017 placed zero orders in 2018. These churned sellers represent lost supply capacity and foregone revenue. The analysis used a CTE with LEFT JOIN anti-pattern to detect absence in 2018.

**Recommendation:** Flag churned sellers for targeted re-engagement campaigns. Offer incentives such as reduced fees or dedicated account manager support.

## 9. Advanced Analytics

### 9.1 RFM Customer Segmentation

RFM analysis segments customers by three dimensions:

- **Recency (R)** — Days since the customer's last purchase (lower = more recent)
- **Frequency (F)** — Number of unique orders placed (higher = more loyal)
- **Monetary (M)** — Total spend across all orders (higher = more valuable)

| Customer ID | Frequency | Monetary |
|---|---|---|
| 0a0a92112bd4c708ca5fde... | 1 | R$ 13,440.00 |
| 763c8b1c9c68a0229c42c9... | 1 | R$ 7,160.00 |
| da122df9eeddfedc1dc1f5... | 2 | R$ 7,388.00 |
| dc4802a71eae9be1dd28f5... | 1 | R$ 6,735.00 |
| 459bef486812aa25204be0... | 1 | R$ 6,729.00 |
| 4007669dec559734d6f53e... | 1 | R$ 5,934.60 |

**Finding:** Most high-monetary customers have Frequency = 1. These high-value single-purchase customers are the prime target for retention campaigns.

### 9.2 Month-over-Month Revenue Growth

MoM analysis was performed using the `LAG()` window function over monthly revenue aggregated with `DATE_TRUNC('month', ...)`. The analysis covers 2016–2018. Key observations: revenue is highly seasonal; March and November are strong recovery months following slow periods.

### 9.3 Seller Churn Detection

Using a two-CTE approach, sellers active in 2017 but absent in 2018 were identified. The pattern uses a LEFT JOIN where the 2018-side IS NULL, which is a standard industry technique for churn detection in transactional data.

## 10. SQL Concepts Applied

All analysis was performed in PostgreSQL. The following concepts were used across the 21 business questions:

- `GROUP BY`, `ORDER BY`, `HAVING`, `WHERE` filtering
- Multi-table `JOIN` (INNER, LEFT) across up to 4 tables
- `COUNT(DISTINCT ...)` for deduplication
- `EXTRACT()` and `DATE_TRUNC()` for time-based analysis
- Window functions: `LAG()`, `SUM() OVER()`
- Common Table Expressions (CTEs) via `WITH` clause
- `FILTER (WHERE ...)` for conditional aggregation
- `ROUND()`, `TRIM()`, `TO_CHAR()` for formatting
- Subquery-based anti-join pattern for churn detection

## 11. Key Insights Summary

1. **São Paulo dominates.** Accounts for R$5.20M (38%) of revenue and 40K+ customers. A single-state dependency risk.
2. **Credit card is king.** 75.24% of all orders paid via credit card. Optimising this checkout flow directly impacts revenue.
3. **Delivery speed drives reviews.** 5-star reviews average 10.62 delivery days vs. 21.25 days for 1-star. Speed is the #1 satisfaction driver.
4. **Northern states suffer.** Roraima (29.3 days) and Amapá (27.2 days) are outliers requiring regional logistics partnerships.
5. **Critical loyalty gap.** 3.12% repeat rate — 96.88% of customers never return. This is the biggest growth opportunity.
6. **Volume inflates late delivery counts.** Monitor rate (%), not absolutes, for Bed Bath Table and other high-volume categories.
7. **Monday–Tuesday are peak order days.** Ideal for flash sales and campaigns.
8. **Top seller concentration.** Top 10 sellers = 14% of revenue. Diversify the seller base to reduce supply risk.

## 12. Conclusions & Recommendations

| Area | Finding | Recommendation |
|---|---|---|
| Delivery | Northern states: 25–30 day avg | Partner with regional logistics providers |
| Retention | 3.12% loyalty rate | Launch rewards program + post-purchase emails |
| Revenue | SP = 38% of total | Diversify into Rio de Janeiro and Minas Gerais |
| Payments | 75% credit card share | Optimise UX; offer instalment incentives |
| Sellers | Top 10 = 14% of revenue | Diversify base; re-engage churned sellers |
| Marketing | Mon–Tue peak order days | Schedule promotions on peak days |
| Satisfaction | Delivery time = review score | Speed investment = direct ROI on ratings |
| Loyalty | 96.88% single-purchase rate | RFM-based VIP campaigns for top spenders |

## Technical Notes

- All queries written and executed in **PostgreSQL**.
- `customer_unique_id` used (not `customer_id`) to avoid double-counting customers with multiple orders.
- Revenue = product price only; freight charges excluded unless stated.
- "Late delivery" defined as: `order_delivered_customer_date > order_estimated_delivery_date`.
- MoM growth excludes the first month (no prior-month baseline).
- RFM recency calculated relative to the latest order date in the dataset.
- Dataset source: Kaggle — Olist Brazilian E-Commerce (CC BY-NC-SA 4.0).

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